

TASKFORGE | PROOF OF CONCEPT

Task Management Platform

POC detail and high-level architecture for the TaskForge application.

Prepared: 10 September 2026

Technology: .NET MAUI, Shell, CommunityToolkit.Mvvm, dependency injection, and SQLite.

Scope: users, tasks, projects, notifications, documents, dashboard, profile, and device capability demonstrations.

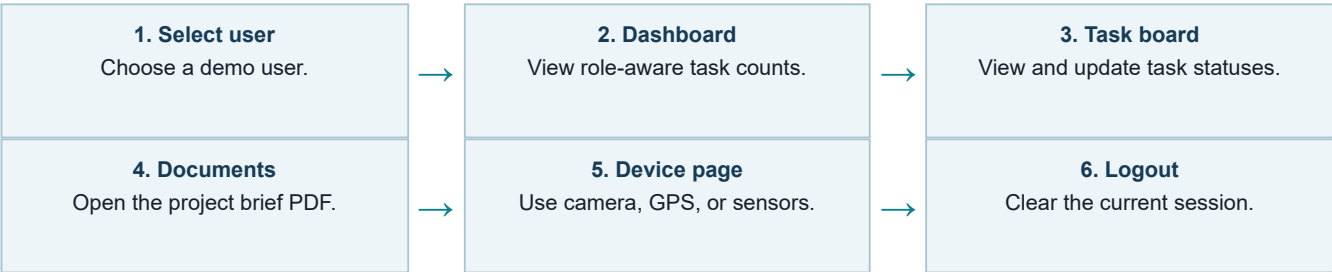
1. POC Overview

TaskForge is a cross-platform task and project management application. The POC demonstrates local data storage, role-aware task views, MVVM screens, Shell navigation, and device capability access.

Area	Current implementation
User access	Simple demo-user selection with in-memory session state and logout.
Task management	Create, assign, view, filter, and update task status.
Dashboard	Shows real task counts from SQLite. Managers and administrators see all tasks; employees see assigned tasks.
Documents	Displays the bundled project brief and opens the PDF using the platform launcher.
Device capabilities	UI for camera capture, GPS lookup, and accelerometer readings with permission feedback.
Storage	Local SQLite database at <code>FileSystem.AppDataDirectory/taskforge.db3</code> .

POC boundary: TaskForge currently has no backend API, cloud synchronization, password authentication, or multi-device data sharing.

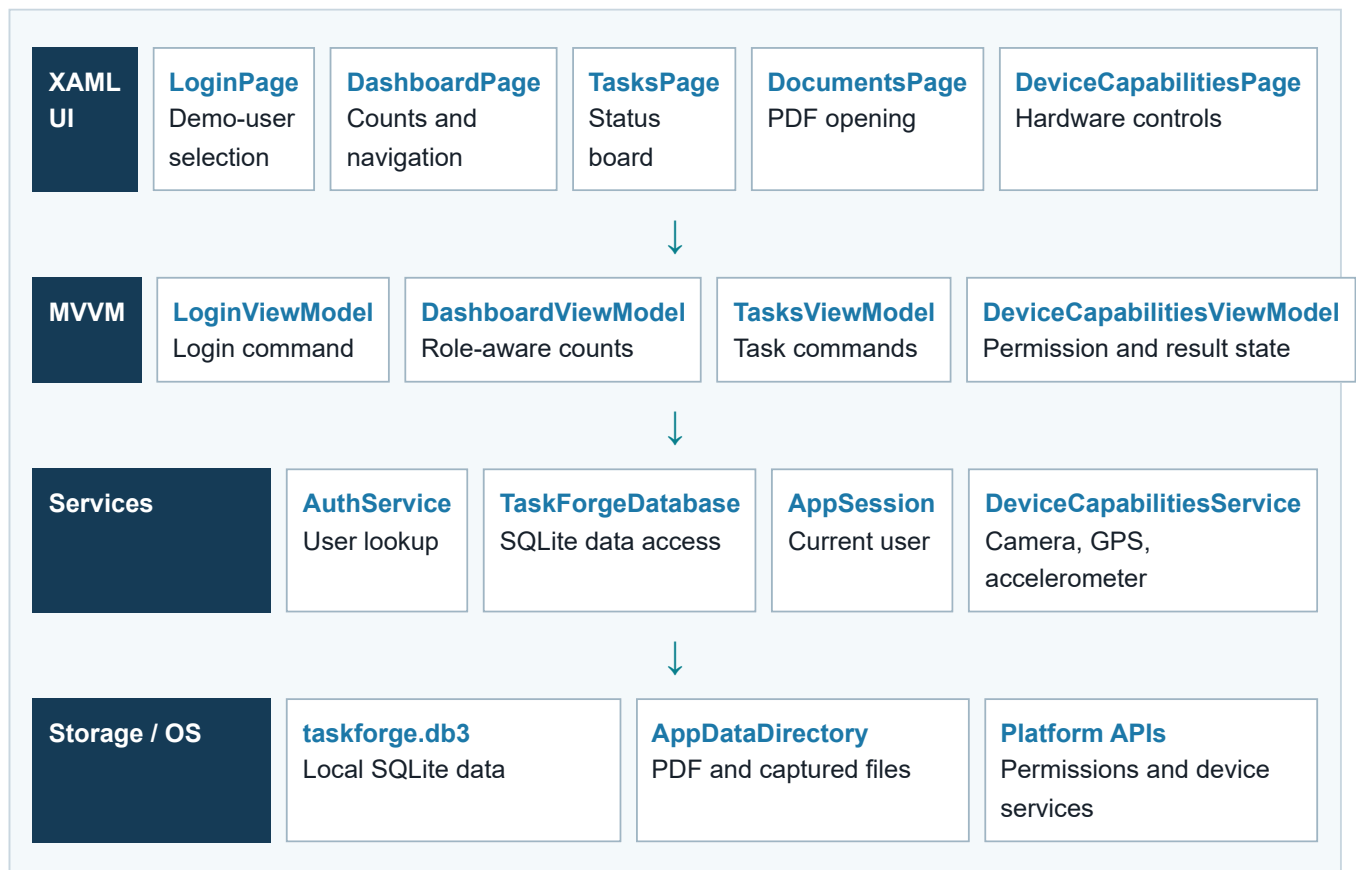
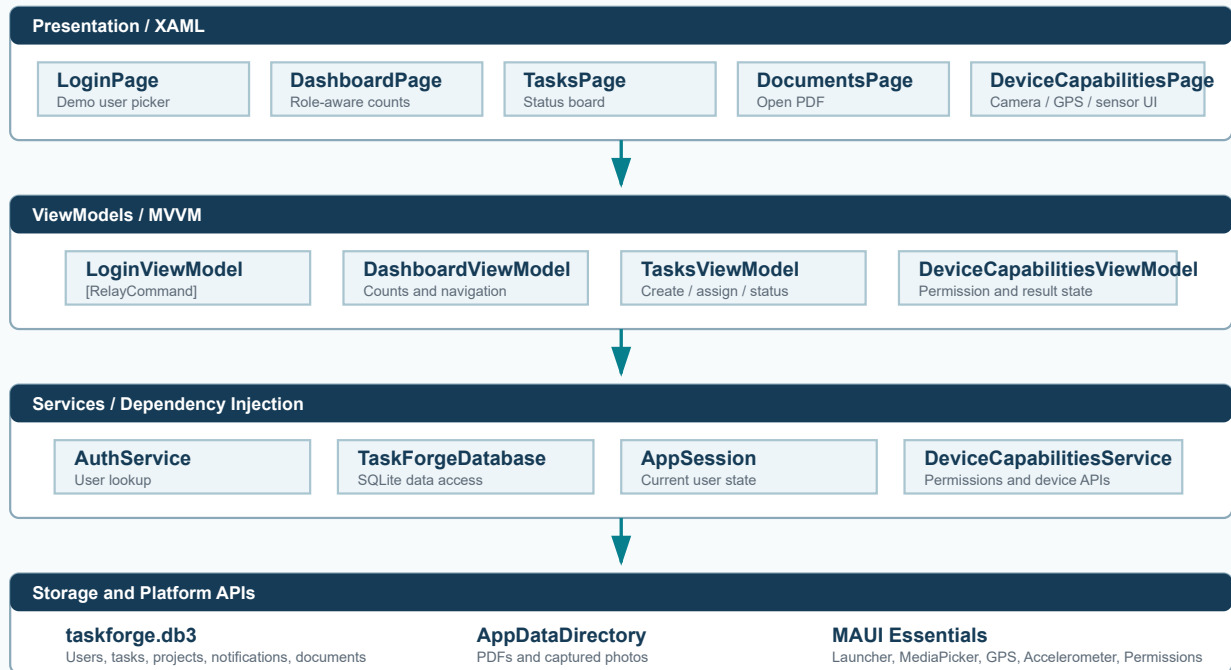
2. Main User Flow



3. High-Level Architecture

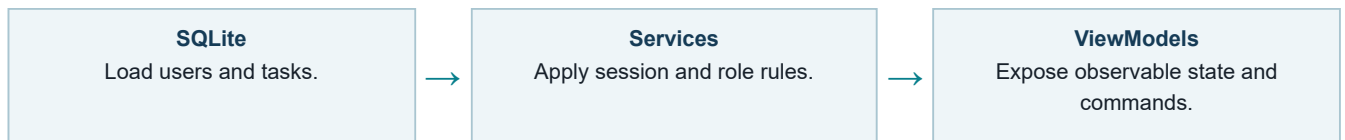
TaskForge High-Level Architecture

Current local POC: .NET MAUI + Shell + MVVM + SQLite



Dependency injection: MauiProgram registers services, ViewModels, and pages. **Navigation:** AppShell defines Login, Dashboard, Tasks, Projects, Team, Notifications, Documents, Profile, and Device Capabilities routes.

Data flow



4. Task and Document Design

Feature	Behavior
Task counts	Administrators, project managers, and team leads use all task rows. Employees use rows assigned to their user ID.
Task board	Shows title, description, due date, visible status, and status picker.
Task creation	Authorized users create tasks through AddTaskCommand, which inserts into SQLite and creates a notification.
Status updates	Picker changes update the TaskItem status in SQLite.
PDF document	The bundled project-brief.pdf is copied to app storage and opened with Launcher.OpenAsync.
Notifications	Assignment notifications are stored and can be marked read.

5. Device Capability UI

Capability	Screen behavior
Camera	Requests camera permission, captures a photo, saves it in app storage, and displays the result.
GPS	Requests location permission and displays latitude and longitude.
Accelerometer	Starts and stops monitoring and displays live X, Y, and Z values.
Platform styling	Android and iOS Entry handler mapper examples are registered in MauiProgram.

6. Acceptance Criteria

- Every demo user receives the correct role-aware task counts.
- Open tasks displays the complete permitted task list.
- Each task row shows its current status.
- The project brief PDF opens from the Documents page.
- Camera, GPS, and accelerometer controls show permission and result states.
- Logout clears the in-memory session.
- The Windows target builds successfully.

7. Production Next Steps

1. Replace demo-user selection with a backend identity service.
2. Move tasks and documents to an API-backed database.
3. Add token authentication, server-side authorization, and audit logging.
4. Persist captured photos and locations against task records.
5. Add offline synchronization and conflict resolution.
6. Add automated tests for task counts, task status updates, PDF opening, and device permissions.

8. Implementation Reference

Area	Files
Database	FieldPro/Data/TaskForgeDatabase.cs FieldPro/Services/ITaskForgeDataService.cs
Task board	FieldPro/ViewModels/TasksViewModel.cs FieldPro/Views/TasksPage.xaml
Dashboard	FieldPro/ViewModels/DashboardViewModel.cs FieldPro/Views/DashboardPage.xaml
Documents	FieldPro/Views/DocumentsPage.xaml.cs FieldPro/Resources/Raw/project-brief.pdf
Device UI	FieldPro/ViewModels/DeviceCapabilitiesViewModel.cs FieldPro/Views/DeviceCapabilitiesPage.xaml
Startup	FieldPro/MauiProgram.cs FieldPro/AppShell.xaml